

Alex Evans

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Education

University of Oxford Oct. 2025 – Current
DPhil Materials

- **Research:** Exploring the application of projection and energy-guided sampling in the de novo design of virus-like particles.

Imperial College London Oct. 2023 – Oct. 2024
MRes Machine Learning in the Physical Sciences - Distinction

- **Research:** Applied diffusion and flow-based techniques in modelling quantum apparatus used in dark matter and gravitational wave research.

University of Birmingham Oct. 2016 – June 2020
BEng Mechanical Engineering - Upper 2:1

Research

AI Driven European Space Agency Satellite Retrieval June 2024 - Oct. 2025
National Physical Laboratory (NPL)

- Developed a generative AI model for the European Space Agency TRUTHS (Traceable Radiometry Underpinning Terrestrial and Helio-Studies) mission, aiming to calibrate the satellite's onboard HIS (Hyperspectral Imaging Spectrometer) to retrieve solar irradiance from thermal readings to observe the effects of climate change mitigation policy.
- Modelled the encoded posterior through the use of a continuous-time diffusion model with transformer architecture for conditioning, enabling samples of the measurand to be generated and therefore providing reliable uncertainty quantification on predictions.
- Techniques: Diffusion; Key Libraries: JAX

Generative AI for Cold Atom Interferometry Nov. 2023 - Oct. 2024
Imperial College London

- Modelled quantum apparatus used in AION (Atom Interferometer Observatory and Network), a national project across various UK universities aiming to detect ultra-light dark matter and gravitational waves through cold-atom interferometry. This model facilitated the application of Bayesian inference to find the ideal set of parameters for optimal cooling.
- Ran 2^{19} atomic simulations with parameters sampled from a Sobol sequence on the Imperial College London HPC servers, totalling ≈ 22 years of total computational time across 500 concurrent simulation instances.
- Applied normalising flows and diffusion in developing a model to generate cooled atoms for given sets of parameters, with generated data from the diffusion model being indistinguishable from the ground-truth to a random forest classifier.
- Developed novel techniques for normalising flow and diffusion models. For normalising flows, by initially denoising the data over a set number of epochs, manifold overfitting was found to be significantly reduced. Similarly for diffusion models, by stabilising an approximation of the trace of the Hessian (Jacobian of the score), overfitting was found to be mitigated.
- Applied Bayesian optimisation on the simulation to find the optimal set of parameters to maximise an objective function derived from probability of transmission, average velocity and atom dispersion.
- Techniques: Diffusion, Normalising Flows, Deep Neural Networks, Decision Trees, Bayesian Optimisation; Key Libraries: JAX, PyTorch, HyperOpt, XGBoost

Solving Inverse Problems in Medical Imaging with Diffusion Nov. 2023 - April 2024
Imperial College London

- Extended literature review on the application of diffusion and novel samplers in denoising medical images; the resulting paper being awarded the highest grade in the cohort.

- Covered the theory of score based approaches, from Hyvärinen score matching to Denoising Diffusion Probabilistic Models and transformer architectures, extending to novel Tweedie-based samplers for solving inverse problems and their potential application in medical imaging.

AI for Autonomous Marine Vessels

Jan. 2024 - June 2024

National Physical Laboratory (NPL)

- Collaborated with the UK Hydrographic Office on a project to develop a standardised language to be interpreted by autonomous marine vessels, making significant strides towards a machine-interpretable model of the S100 series of navigational charts.
- Developed an LLM workflow to translate tags from the S100 series of navigational charts into machine-readable format, with XML validation against an XSD schema.
- Techniques: Large Language Models; Key Libraries: Ollama

Employment

Scientist

Teddington, London

National Physical Laboratory (NPL)

Jan. 2024 – Oct. 2025

- Collaborated with the Earth Observation and Climate group as a part of the TRUTHS (Traceable Radiometry Underpinning Terrestrial and Helio-Studies) mission to develop a generative AI model to calibrate a European Space Agency satellite to observe effects of climate change mitigation strategies.
- Worked with the Informatics group on both HADO (High Intensity Autonomous Drone Operations) and MAAT (Maritime Autonomy Assurance Testbed), assuring the development and deployment of autonomous security drones at Heathrow Airport and autonomous vessels.

Data Analyst

Teddington, London

National Physical Laboratory (NPL)

Nov. 2021 – Jan. 2024

- Utilised Azure cloud platform pipelines to create a unified SQL database of data from across the organisation, enabling finance and project management data to interact and therefore significantly improving the capability and quality of organisational reporting.

Awards

STEM for Britain Finalist

March 2025

Department for Science, Innovation and Technology

- Selected as a finalist for STEM for Britain 2025, presenting research from Imperial at parliament to MPs and judges from academia.

Academic Achievement

Oct. 2024

Imperial College London

- Achieved top 3 final grade in cohort; rank 1 in Statistics for Experimental Physics and Self-Study Project on score-matching and diffusion in medical imaging.

Presentations

Science and Technology Advisory Council (STAC)

June 2025

National Physical Laboratory (NPL)

- Due to present poster from parliament to STAC.

RAEng Africa Prize

March 2025

National Physical Laboratory (NPL)

- Presented poster on Generative AI for Cold Atom Interferometry to nominees for the RAEng Africa Prize visiting NPL.

STEM for Britain 2025

March 2025

UK Parliament

- Presented poster at parliament on Generative AI for Cold Atom Interferometry to MPs and judges from academia.